National College of Computer Studies

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Report on

5G Network

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Abstract

The rise of 5G networks is examined in depth in this research study along with the opportunities, advantages, and difficulties it presents. 5G networks offers faster and more reliable internet connections, along with low latency, high bandwidth, and ability to connect more devices simultaneously. The paper examines the prospective benefits of 5G networks, such as the ability to enable new apps and services, support the Internet of Things, and deploy network slicing. The advantages of 5G networks, such as quicker and more reliable internet connections, the development of new business models, and the deployment of edge computing, are also explored. There is also a discussion of the issues involved with 5G networks, such as infrastructure investment, security concerns, potential health risks, and the impact on employment opportunities.

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1. Introduction/Background

5G Technology is an abbreviation for fifth generation mobile technology. Beyond the upcoming 4G standards, 5G denotes the next major phase of mobile telecommunication standards. 5G technology provides services such as product engineering, documentation, and electronic transaction assistance, among other things. As customers become more aware of mobile phone technology, they will look for a good bundle that includes all of the modern capabilities that a cellular phone may have.

Figure 1: 5G network

As a result, the quest for new technology has always been the primary goal of the world's leading cell phone companies in order to out-innovate their competitors. A 5G-based telecommunications network would ideally address the issues that a 4G model would offer once it became widely used.

Wireless systems that use orthogonal frequency division multiplexing (OFDM) to provide large area coverage, high capacity at millimeter waves (10 mm to 1 mm), and a data rate of 20 Mbps at distances of up to 2 km. The millimeter-wave spectrum is the most efficient way to address the recent spike in wireless Internet consumption. These specs can support wireless world wide web (WWWW) applications. The WWWW enables a highly flexible network (flexible channel bandwidth between 5 and 20 MHz, optimally up to 40 MHz). Users of 5G technology can download a whole film, including 3D movies, to their tablets or laptops, as well as play games and access remote medical services. Piconet and Bluetooth technologies will become obsolete with the introduction of 5G. The 5G mobile phones would be similar to tablet PCs, allowing you to view TV channels in HD quality without interruption.

2. History of 5G network

a. 1G Network: Nippon Telephone and Telegraph Company (NTT) launched Japan's first generation mobile network in Tokyo in 1979. Due to technological restrictions, this system utilized analog signals and had numerous drawbacks.

b. 2G Network: It included a new digital technology for wireless transmission known as the Global System for Mobile Communication (GSM). GSM technology later established the foundation for future wireless standard development. This standard was capable of supporting data rates ranging from 14.4 to 64kbps (maximum), which is sufficient for SMS and email services.

c. 3G Network: With the launch of UMTS - Universal Mobile Terrestrial / Telecommunication Systems – third-generation mobile communication began. UMTS has a data rate of 384kbps and, for the first time on mobile devices, it supports video calling. Smartphones became popular all over the world following the advent of the 3G mobile communication system.

d. 4G Network: 4G systems are an improved version of IEEE's 3G networks, with better data rates and the ability to handle more complex multimedia services. In fourth-generation systems, LTE and LTE advanced wireless technology are used. Furthermore, it is backward compatible with previous versions, allowing for easier deployment and upgrade of LTE and LTE advanced networks.

e. 5G Network: Earlier deployments of the 5G network will operate in both non-standalone and standalone modes. In non-standalone mode, both LTE and 5G-NR spectrum will be used concurrently. In non-standalone mode, control signaling will be linked to the LTE core network.

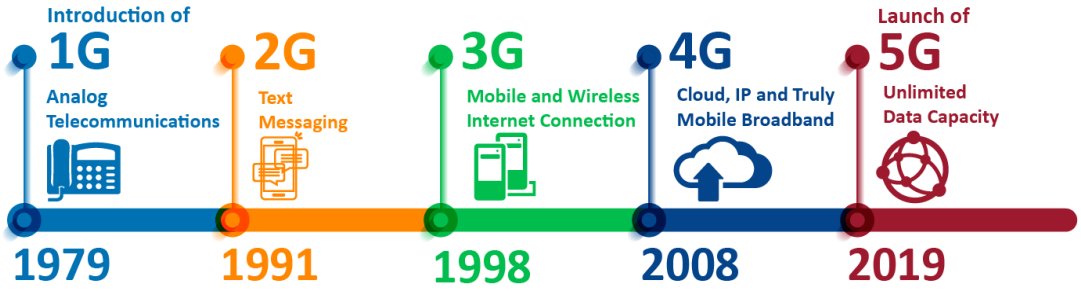

Figure 2 History Of 5G network

2.1. Difference between 4G and 5G

4G and 5G are both wireless communication technologies, but 5G offers several key improvements over 4G:

Speed: 5G is designed to be much faster than 4G, with peak speeds up to 20 times faster.

Latency: 5G has significantly lower latency than 4G, which means there is less delay between sending and receiving data.

Capacity: 5G can handle more devices and more data than 4G, which makes it better suited for supporting the increasing number of internet-connected devices.

Spectrum: 5G uses higher frequency bands than 4G, which allows for more data to be transmitted at once.

Energy efficiency: 5G is designed to be more energy efficient than 4G, which could lead to longer battery life for devices and lower overall energy consumption.

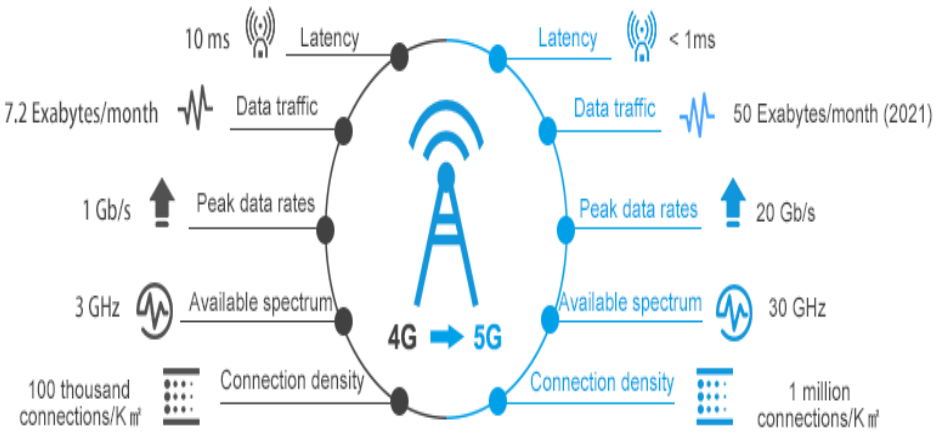

Figure 3: Comparing 4G and 5G

Table 2: Difference between 4G and 5G

4G Technology	**5G Technology**
The maximum upload rate of 4G technology is 500 Mbps	While the maximum upload rate of 5G technology is 1.25 Gbps
The maximum download rate of 4G technology is 1 Gbps	While the maximum download rate of 5G technology is 2.5 Gbps.
The latency of 4G technology is about 50 ms	While the latency of 5G technology is about 1 ms
4G offers CDMA	While 5G offers OFDM, BDMA
4G can't differentiate between fixed and mobile devices	While 5G has the capability to differentiate between fixed and mobile devices. It uses cognitive radio techniques to identify each device and offer the most appropriate delivery channel.
4G has the advantages of high-speed handoffs, global mobility	While 5G has the advantages of extremely high speeds, low latency

3. Benefits of 5G Networks

The deployment of 5G networks will bring several benefits to individuals, businesses, and society as a whole. Firstly, it will enable faster and more reliable internet connections, which will enhance the user experience and enable the development of new applications and services. Secondly, the high bandwidth of 5G networks will enable the connection of a vast number of devices simultaneously, which is essential for the development of smart cities, smart homes, and the Internet of Things. Thirdly, 5G networks will enable the deployment of edge computing, which will enable the processing of data closer to the source, reducing latency and improving the performance of applications and services. Finally, 5G networks will enable the development of new business models.

a. Bi-directional huge bandwidth shaping with high resolution.
b. Technology that connects all networks onto a single platform.
c. More efficient and effective..
d. Technically sound to support a variety of services (including a private network).
e. It is possible to provide global connectivity that is uniform, uninterrupted, and consistent.

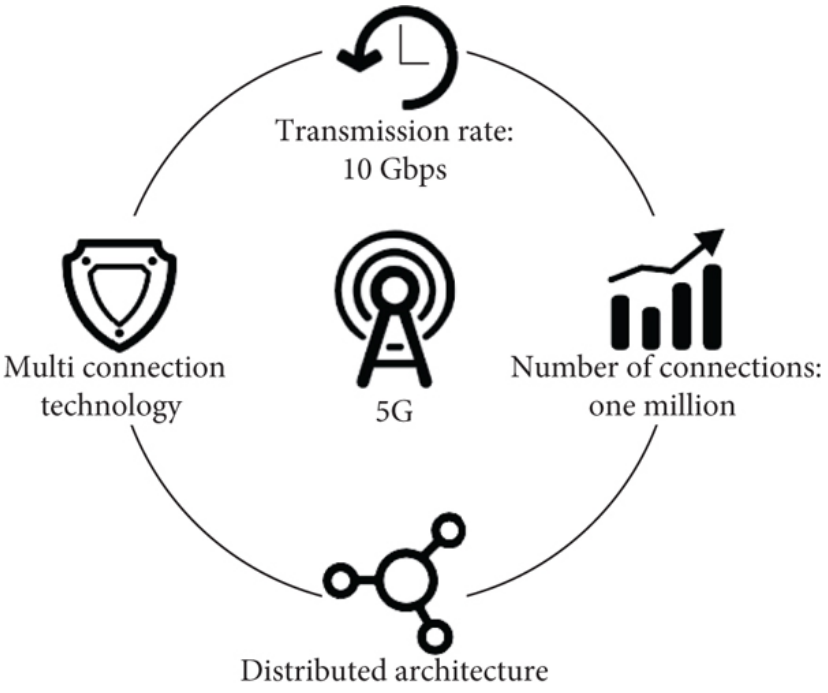

Figure 4: Benefits of 5G

4. Prospects of 5G network

The prospects of 5G networks are significant, offering numerous benefits that were not possible with previous generations of mobile telecommunications technology. One of the most significant prospects of 5G networks is the ability to support new applications and services that require real-time response, such as virtual and augmented reality, remote surgeries, and autonomous vehicles. Additionally, 5G networks offer the ability to connect a massive number of devices simultaneously, which is critical for the development of the Internet of Things. Another prospect of 5G networks is the deployment of network slicing, which allows operators to allocate network resources dynamically based on the needs of specific applications or services

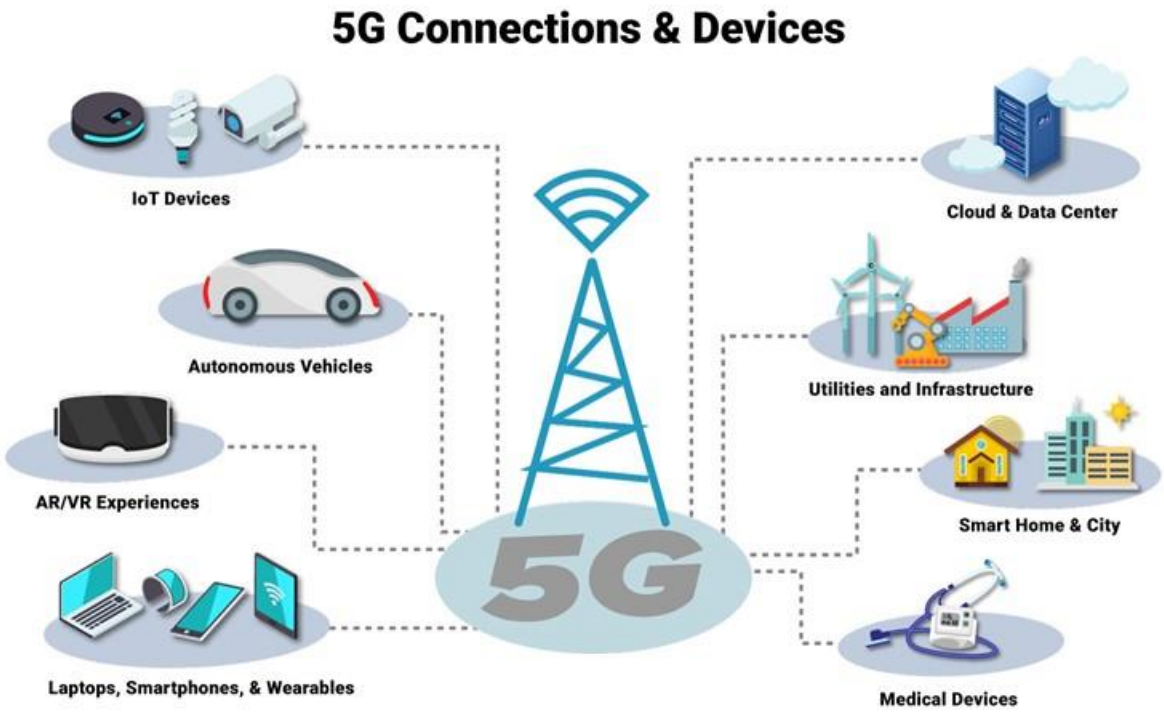

Figure 5: 5G Connections and Devices

5. Disadvantages

Though 5G technology has been researched and conceptualized to solve all radio signal problems and hardships in the mobile world, it has the following shortcomings due to security concerns and a lack of technological advancement in most geographic regions. Many older devices will be incompatible with 5G, necessitating the purchase of a new, more expensive model. Security and privacy concerns have yet to be addressed.

Cost: One of the greatest impediments to the practical deployment of 5G technology is its high cost, particularly when it comes to widespread adoption. For example, 5G subscription plans will almost certainly be prohibitively expensive in the beginning.

Coverage and Radio Frequency Issues: According to some sources, 5G technology will require a frequency of 6 GHz or less. The issue with this frequency spectrum is that it is already congested, and adding 5G connection to the mix may cause data transmission challenges, making it harder to send and receive signals. It is also feasible that 5G network cell users will have worse coverage than 4G models. This means that, although having more capacity, 5G technology will require more cell towers to achieve large-scale 5G network integration. As a result, 5G towers are predicted to offer only 300 meters of outdoor coverage and two meters of internal coverage.

Signal failures are possible: It is quite possible that 5G technology will result in significant losses in the network's millimeter wave, due to challenges such as rain reduction, vegetation losses, penetration issues, and so on. There is no certainty that 5G technology will meet end-user and software developer speed expectations.

5.1. Does 5G affects health?

The current scientific agreement is that there is no evidence that 5G technology is harmful to human health. 5G technology, like previous generations of wireless technology, uses radio waves to transport data, but it runs at higher frequencies and employs more antennas to boost communication throughput and reduce latency. The radio waves utilized by 5G are in the non-ionizing region of the electromagnetic spectrum, which means they lack the energy to ionize atoms or molecules, which can harm human cells.

Fears have been made concerning the potential health impacts of 5G, however scientific research to date does not support these fears. According to the World Health Organization, "considering the very low exposure levels and research results collected to date, there is no convincing scientific evidence that the weak RF signals from base stations and wireless networks cause adverse health effects."

However, it is important to note that research into the health effects of 5G technology is ongoing, and scientists will continue to study the potential effects of radio wave exposure as the technology becomes more widely available.

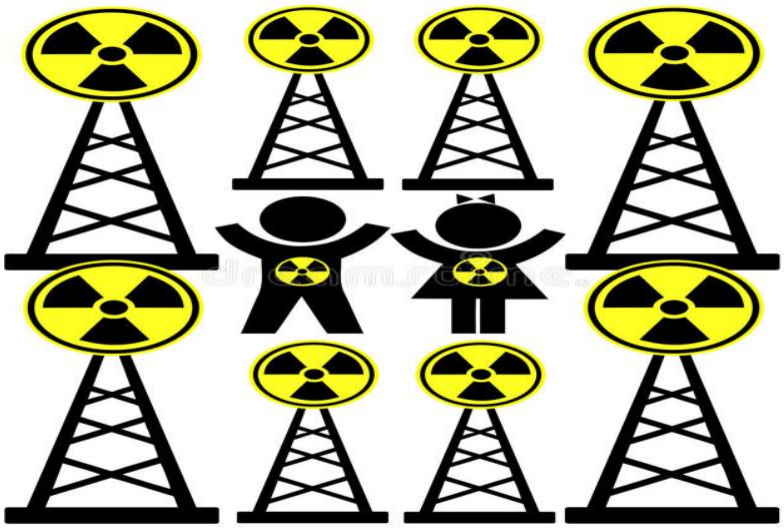

Figure 6: Risk of Health from 5G

6. Challenges of 5G Networks:

a. Spectrum and frequency band availability concerns: Following the widespread adoption of 5G technology, entirely new use cases will emerge. This will lead to an increase in demand for high-frequency bands. However, due to the high cost and scarcity of spectrum, CSPs must develop a compelling business case. Because these spectrums must be purchased from governments through auction, telecom operators must select frequency bands and adjust their 5G network and features accordingly. This could result in greater operational expenditures in order to provide top-tier 5G services with a constrained spectrum range.

b. Methodology for deploying 5G networks: To begin, CSPs must have a clear strategy for implementing 5G network slicing and other arrangements. Second, after deciding on a strategy, their approach to deployment will determine the outcome of the deployment process. CSPs will create their deployment model and approach based on the spectrum networks purchased, which will eventually be required for addressing specific 5G use cases. Furthermore, 5G network implementation challenges include the widespread use of mm-Wave frequencies and 5G small cell towers, which will necessitate a new deployment strategy while adhering to regulations.

c. End-user mobile devices must be upgraded: In the early stages of 5G implementation, the new generation of mobile devices is designed to support 5G bands at least in the sub-6 GHz range. However, in order for 5G network implementation to be successful, a large number of 4G devices must be replaced with 5G-compatible ones.

7. Conclusion:

5G networks are the latest generation of mobile telecommunications technology, which offers faster and more reliable internet connections, low latency, high bandwidth, and the ability to connect more devices simultaneously. The deployment of 5G networks will bring several benefits to individuals, businesses, and society as a whole. However, there are several challenges that need to be addressed, including significant investment in infrastructure, security concerns, potential health risks, and potential impact on employment. Overall, the benefits of 5G networks outweigh the challenges, and the deployment of 5G networks is essential for the development of the digital economy.

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